

Real-Time Coherent OFDR Over Live Networks: From Access to Subsea

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Abstract: We demonstrate how optical frequency domain reflectometry with continuous power probe pulses enables real-time distributed fiber-optic sensing over various network configurations ranging from 1x32 PON to trans-oceanic submarine systems and networks using hollow-core fiber.

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1. Introduction

Interest in fiber sensing, within the optical fiber community, has surged over the last decade. With billions of kilometers of single-mode fiber deployed worldwide, the potential for these networks to serve a dual purpose of both communications and environmental sensing has become more compelling [1]. Broadly speaking, the sensing methods utilizing deployed fiber-optic networks belong in two categories: leveraging existing network signals and those that introduce dedicated fiber sensing instruments and methods into the network infrastructure. In telecommunications, loss monitoring normally relies on Rayleigh backscattering measured with optical time domain reflectometry (OTDR). Similarly, phase-sensitive measurement of Rayleigh backscattering is the foundation for distributed acoustic sensing (DAS), a key technique of the fiber sensing community.

In this paper we review recent developments on coherent fiber optic sensing focusing on integrating fiber sensing methods into telecommunication networks. Central to this work is the recent improvements in real-time optical frequency domain reflectometry (OFDR) sensing for telecom applications. We compare OFDR to OTDR, with particular emphasis on system design from a telecom compatibility perspective. Specifically, we address hybrid applications where live networks perform environmental sensing with negligible impact on the network throughput. We will cover networking architectures from single-span configuration, to passive optical networks (PON) and trans-oceanic submarine networks, as well as emerging transmission systems using anti-resonant hollow-core fiber. In addition to operating on telecom networks, we also seek to utilize telecom-based components such as integrated coherent receivers and DSP implemented on telecom-grade FPGA platforms.

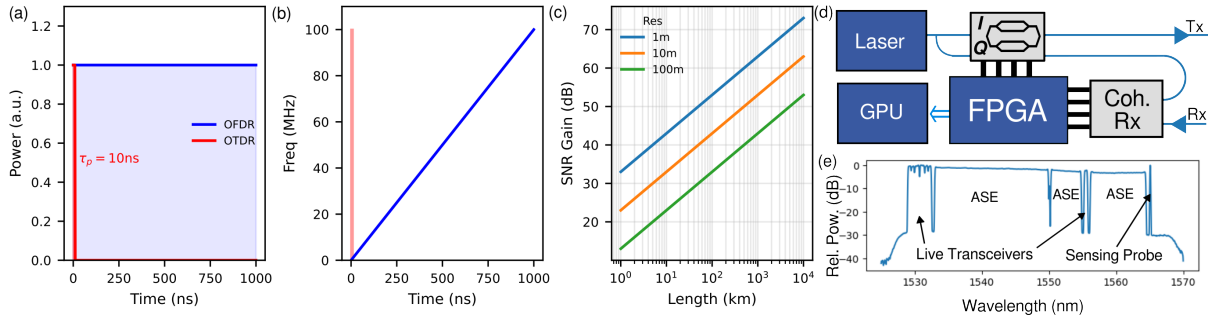


Fig. 1: (a) Comparison of power transmitted using OTDR and OFDR with $\tau_{pulse} = 10\text{ ns}$ and a repetition ratio of $T_{scan} = 1\mu\text{s}$. (b) Comparison of the transmitted frequency for the same setup. (c) Theoretical SNR gain when using OFDR over OTDR for 1, 10 and 100 m spatial resolution for different fiber lengths. The gain exceeds 30 dB for a 100 km system using 10 m spatial resolution. (d) Experimental setup for a coherent OFDR implementation using telecom components. (e) Example of adding a fiber sensing probe into a live submarine network by replacing existing monitoring equipment with a real-time OFDR probe.

2. Optical Time and Frequency Domain Reflectometry Comparison

While phase-sensitive OTDR currently dominates DAS implementations [1], OFDR [2] presents distinct advantages for telecom integrated sensing. To compare these implementations, we analyze the waveforms launched into the fiber. The periodic output fields, denoted as E_{OTDR} and E_{OFDR} , are defined as:

$$E_{OTDR} = \sqrt{P_{\text{Peak}}} \text{rect}\left(\frac{t}{\tau_{pulse}}\right), E_{OFDR} = \sqrt{P_{\text{avg}}} \exp(j\pi\gamma t^2), \quad (1)$$

where P_{Peak} , P_{avg} denote the peak and average pulse power, and $\text{rect}()$ the rectangular window function. The laser sweep parameter $\gamma = B/T_{scan}$, with B being the total bandwidth swept and T_{scan} the duration of the sweep. The spatial resolution limit, Δz for each case is

$$\Delta z_{OTDR} = \left(\frac{c\tau_{pulse}}{2n_{eff}}\right), \Delta z_{OFDR} = \left(\frac{c}{2n_{eff}B}\right), \quad (2)$$

where c is the speed of light and n_{eff} fiber refractive index. Assuming a typical value of $n_{eff} = 1.46$ for single-mode fiber (SMF), a 10 ns OTDR pulse gives a spatial resolution of about 1 m. Achieving the equivalent spatial resolution for an OFDR system requires a bandwidth $B = 1/\tau_{pulse}$ of 100 MHz.

A comparison between the waveforms' power in the time and frequency domains is shown in Fig. 1(a) and (b), respectively. In both cases, because the Rayleigh scattering spreads the delay of the returned light, the time between consecutive pulses for an OTDR T_{rep} , or correspondingly, the sweep duration T_{scan} for an OFDR must be longer than the roundtrip time of the fiber to prevent overlap. We can therefore say that $T_{rep} = T_{scan} \geq \frac{2n_{eff}L}{c}$, with L denoting the fiber length. A critical constraint in telecom networks is the limit on launch power to prevent nonlinear distortions in neighboring channels. As illustrated in Fig. 1(a), we see that an OTDR system is only transmitting light during a small fraction of the time. We can define the ratio E_r between the transmitted energy of the OFDR and OTDR systems according to $E_r = \frac{T_{scan}}{\tau_{pulse}}$, which for an ideal system without any distortions will be directly equivalent to an improvement in signal-to-noise ratio SNR. Fig. 1(c) illustrates the theoretical SNR gain for a spacing of 1, 10 and 100 m as a function of fiber length. As an example, the theoretical SNR gain exceeds 30 dB for a 100 km long fiber and 10 m spatial resolution target, which is typical for deployed DAS systems. While practical factors such as laser phase noise and sweep nonlinearity will reduce the performance [4], as evident from Fig. 1(c), the potential SNR gain is substantial. This makes OFDR a particularly attractive candidate in live networks where the SNR gain can be used to lower the peak power, thereby mitigating nonlinear interference with telecom channels. Furthermore, this gain is essential for sensing in high-loss environments, such as submarine links with high-loss loopback couplers, high split ratio PON architectures, and networks using anti-resonant hollow-core fiber. In all cases, the backscatter level is typically about 35-40 dB less than found in a traditional single-mode fiber (SMF).

3. Coherent OFDR Implementation

The performance of an OFDR system is fundamentally dictated by the quality of its frequency sweep. OFDRs with THz bandwidth are standard for characterizing fibers and components and used in LIDAR [5] and these systems rely on a sweeping laser source to implement the chirp and the bandwidth can easily reach exceeding 10 THz, covering both the C- and L-bands. However, telecom-integrated sensing has different requirements, and as outlined in section 2, GHz-level bandwidths are sufficient for achieving $\approx$ m-level spatial resolutions. Therefore, an OFDR-based DAS system can be implemented using acousto-optic modulators [2] or, more recently, with polarization-diverse IQ modulators [6]. The latter are particularly advantageous as they enable full-field waveform control, supporting arbitrary waveforms and advanced DSP to improve system performance [7].

Figure 1(d) illustrates a software-defined sensing system leveraging standard telecom components. The combination of telecom IQ modulator and a polarization-diverse coherent receiver enables full-field control and supports the use of DSP in both the transmitter and receiver to enhance performance. While telecom components can have bandwidths reaching >100 GHz, they are typically not optimized for sensing applications, which can cause unwanted performance degradations. Because the sensing bandwidth requirements are significantly lower than those of high-speed coherent transceivers, processing can be implemented using FPGAs instead of ASICs. This architecture facilitates rapid prototyping and adjustment and updating of field-installed systems. In addition, GPU-based real-time processing can significantly speed up development and enable complementary processing to the FPGA.

When performing sensing over a live network, the fiber sensing signals must be added without distorting existing channels. While the OFDR design is highly suitable for co-existence, networks need to be adjusted to allow for real-time monitoring. In subsea, dedicated monitoring channels are already in place, and the OFDR can therefore seamlessly replace existing monitoring tools and add new capabilities [7]. An example of such implementation for a subsea cable is shown in Fig. 1(e), where the sensing probe is placed in the monitoring channel on the edge of the spectrum. In terrestrial networks, OTDRs are used frequently and OFDR can simply replace existing systems to enable real-time monitoring. Another option involves adding DAS capability to the optical supervisory channel (OSC), enabling per-span DAS while using the supervisory network for data movement and network control [11]. This also provides a direct path for leveraging fiber sensing in optical network monitoring and control systems.

4. Exemplary use cases

An example demonstrating three distinct use cases is illustrated in Fig. 2. As outlined previously, the constant power nature of the OFDR implementation makes it highly suitable for use in live networks.

In the first example, an operational submarine cable is turned into a distributed deep-ocean seismic array, as shown in Fig. 2(a) [6]. Second, OFDR significantly outperforms OTDR in the characterization of fibers with ultra-low backscattering, such as anti-resonant hollow-core fibers [8]. While these can achieve a propagation loss below that of SMF, they have about 40 dB weaker backscatter. By using OFDR optimized for ultra-high dynamic range, the fiber can be fully characterized, and the loss profile measured with a spatial resolution on the order of 10 m [9].

Finally, the characterization of a PON system is shown in Fig. 2(c). Here, a 1:32 splitter was used, leading to a roundtrip loss of about 35 dB in addition to the fiber propagation loss. As seen in Fig. 2(c), about 70 dB dynamic range is required to resolve the entire network. Using OFDR, the entire network can be resolved with a spatial resolution of <10 m in less than a second [10]. The SNR gain from OFDR is essential to ensure real-time characterization without requiring a prohibitively large launch power, which causes nonlinear penalties and traffic interruptions [10].

While only a small selection of use-cases are shown in Fig. 2, they all highlight two foundational benefits of using OFDR for fiber sensing using live telecom networks. First, the constant power probe signal avoids nonlinear distortion, which is fundamentally enabling co-existence. Second, the large SNR gain from OFDR enables real-time characterization over a broad range of use-cases. For instance, submarine cable monitoring can now be performed in real-time if OFDR [7] is used. State-of-the-art OTDR methods and equivalent measurements take typically several hours.

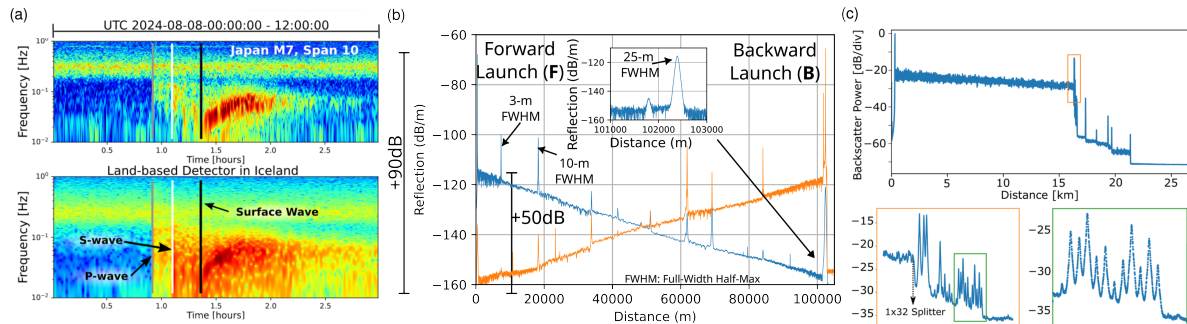


Fig. 2: (a) Use of OFDR for turning an operational submarine cable into a distributed seismic array by integrating the strain between consecutive repeaters. Shows the signal recorded using OFDR and bottom a reference signal from a seismic instrument in a borehole [6]. (b) OFDR with >90 dB dynamic range to characterize anti-resonant hollow-core fiber. The Rayleigh backscatter power level ≈ 40 dB weaker than in SMF and OFDR is used to maximize the SNR [9]. (c) Characterization of an experimental PON setup using a 1x32 splitter. Leveraging OFDR, the entire system can be characterized in <1 s while maintaining a low enough power level to ensure penalty-free operation next to live telecom channels [10].

5. Conclusion

We show that coherent OFDR provides a robust framework for integrating real-time environmental sensing into live telecommunication networks. By leveraging continuous power probe pulses, this approach overcomes the peak power limitations inherent to traditional OTDR, delivering significant SNR gains while minimizing nonlinear interference with co-existing data channels. We highlighted the versatility of this method across diverse network architectures—ranging from high-loss PON configurations and emerging anti-resonant hollow-core fibers to trans-oceanic submarine links, thus confirming that high-resolution sensing is achievable without compromising network throughput. Ultimately, these advancements enable a dual-purpose optical infrastructure where global communication networks seamlessly double as planetary-scale environmental sensors.

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References

1. E. Ip *et al.*, "Distributed fiber sensor network using telecom cables as sensing media...", JOCN, vol. 14, no. 1, A61–A68, 2022
2. O. H. Waagaard *et al.*, "Real-time low noise distributed acoustic sensing...", OSA continuum, vol. 4, no. 2, pp. 688–701, 2021.
3. Z. Ding *et al.*, "Ultimate Spatial Resolution Realisation in Optical Frequency Domain Reflectometry...", Sensors (MDPI), 2021.
4. Z. Ding *et al.*, "Compensation of laser frequency tuning nonlinearity of a long range OFDR using deskew...", Opt. Exp., 21(3), 2013.
5. E. D. Moore, "Advances in swept-wavelength interferometry for precision ...", Diss. University of Colorado at Boulder, 2011.
6. M. Mazur *et al.*, "Global seismic monitoring using operational subsea cable", Proc. ECOC paper Th3B.7, 2024
7. M. Mazur *et al.*, "Real-Time In-Line Coherent Distributed Sensing over a Legacy Submarine Cable", Proc. OFC paper Th4B.8, 2024.
8. M. Petrovich *et al.*, "Broadband optical fibre with an attenuation lower than 0.1 decibel per...", Nat. Phot. 19, 1203–1208 (2025)
9. N. K. Fontaine *et al.*, "Ultra-high resolution and long-range OFDRs for characterizing...", Proc. OFC paper Th4D.6, 2025.
10. P. Iannone *et al.*, "Real-Time, 1-m Resolution Measurement of the Optical Distribution Network...", Proc. OFC paper Th1K, 2026
11. M. Mazur *et al.*, "Coherent Optical Supervisory Channel Enabling Per-Span Real-Time Gigabit/s Transmission...", Proc. ECOC 2024.